

Task 2

What the program does

Imagine two friends, ev1 and ev2, playing catch with a ball. Every time one of them catches the ball, they add 1 to a number written on it, then throw it to the other friend. The game stops when the number on the ball goes above 10.

Starting up

When the program first runs:

- Two promises are created and immediately finished. Think of them as two "ready" signals that are already switched on
- ev1 and ev2 are told what to do when they receive the ball
- ev2 gets the ball first with the number 0 on it
- The program finishes its setup and steps aside, letting the event loop take over

The cycle

The event loop is like a referee watching the game. It keeps doing this on repeat:

1. ev2 (or ev1) catches the ball and shouts out the number. This is the console.log
2. It schedules the next throw using .then()
3. foo() looks at the number on the ball: even number means throw to ev1, odd number means throw to ev2; and adds 1

What gets printed

```
data 0 received by ev2
data 1 received by ev1
data 2 received by ev2
data 3 received by ev1
data 4 received by ev2
data 5 received by ev1
data 6 received by ev2
data 7 received by ev1
data 8 received by ev2
data 9 received by ev1
data 10 received by ev2
data 11 received by ev1
```

It alternates every line because even numbers always go to ev1 and odd numbers always go to ev2.

Does it end?

Yes. When the number reaches 11, foo() sees that $11 > 10$ and stops. It doesn't throw the ball anymore. Both queues are empty and the game is over.

```
(base) natsim#NatBook-Pro stacktable %
```